SkillClaw

Collective Skill Evolution for Multi-User LLM Agent Ecosystems

Let Skills Evolve Collectively with Agentic Evolver

arxiv.org/abs/2604.08377

Agent Skills Remain Static After Deployment

Users Rediscover Solutions Independently

STATIC PARADIGM (Current)

- Skills manually installed from centralized hub
- Solutions discovered during interaction don't persist
- Same failure patterns rediscovered by every user
- No system-level knowledge accumulation

EVOLVING PARADIGM (SkillClaw)

- Skills evolve from runtime interaction evidence
- Cross-user trajectories aggregated as shared signal
- Agentic evolver proposes and verifies mutations
- Collective knowledge improves the whole ecosystem

Existing approaches: Memory-based methods are instance-specific; skill-based methods treat libraries as static resources.

Closed-Loop Pipeline: Multi-User Interaction → Shared Skill Evolution

EVIDENCE

Analyze trajectories to identify behavioral patterns

→

ATTRIBUTION

Diagnose root causes of failures and successes

→

EVOLUTION

Propose skill mutations — refine or create new

→

DISTRIBUTION

Verify, merge into shared repo, sync to all agents

← Continuous closed loop — updated skills generate new trajectories →

MULTI-USER INTERACTION LAYER

User A (OpenClaw)	User B (CoPaw)	User C (IronClaw)	User D (NemoClaw)	User N (ZeroClaw)

Agentic Evolver: Evidence → Attribution → Evolution

Open-ended agent reasoning over interaction evidence — not predefined update rules

1

Evidence

Analyze structured session trajectories that preserve full action–feedback causal chains. Identify behavioral patterns across grouped skill-specific interactions.

2

Attribution

Diagnose root causes of failures and successes. Separate generalizable improvements from idiosyncratic fixes using cross-user signal diversity.

3

Evolution

Propose skill mutations — refining existing skills or creating new ones. Updated skills are verified before being merged into the shared repository.

The evolution process is driven by an autonomous agent performing open-ended reasoning — not predefined update rules.

Cross-User Trajectory Aggregation Exposes Skill Behavioral Boundaries

Why Multi-User Signal Matters

- Different users exercise the same skill under diverse contexts
- Produces complementary views of a skill's behavioral boundary
- Reveals both conditions where a skill works and where it breaks
- A single user rarely generates enough signal to separate generalizable improvement from idiosyncratic

Collective signal > Individual signal

Compatible Claw-Style Systems

OpenClaw	CoPaw
IronClaw	PicoClaw
ZeroClaw	NanoClaw
NemoClaw	

across all systems and grouped by referenced skill

Case Study: Multimodal Creative Pipeline — 6-Day Evolution Log

Only 1 of 6 candidates accepted — conservative verification prevents regressions

Day	Candidate Skill	Skill Function	Result	Action
1	validate-tmp-workspace-inputs	Check /tmp_workspace inputs & env setup	✓ **Accept**	Upgrade to new best pool
2	multimodal-input-validation-and-setup	Multimodal input validation & output init	✗ **Reject**	Keep current best pool
3	multimodal-creative-task-pipeline	Extract from PDF/video/image	✗ **Reject**	Keep current best pool
4	multimodal-creative-task-pipeline (impr.)	Added image classification, visual gen, validation	✗ **Reject**	Keep current best pool
5	multimodal-creative + validate-required-input-files	Pipeline + per-file fail-fast validation	✗ **Reject**	Keep current best pool
6	multimodal-creative-task-pipeline (cand.)	Extended PDF-to-poster / document-to-visual	✗ **Reject**	Not admitted to next cycle

Accepted: 1 / 6 | Rejected: 5 / 6

Case Study: Git Push Auth-Fallback — Iterative Refinement Over 6 Days

3 of 6 days yielded accepted updates — successful refinement before reaching plateau

Day	Candidate Skill	Change Summary	Result	Action
1	git-push-with-auth-fallback	Safe fallback instead of blocking on push failure	✓ Accept	Add to Safety best pool
2	git-push-with-auth-fallback	Unified patch/bundle filenames, reduced inconsistency	✓ Accept	Keep updated best pool
3	git-push-with-auth-fallback + git-clone-to-directory	Auth-alternative paths & secrets audit; fixed subshell	✓ Accept	Keep current best pool
4	(none — retest)	Same-pool retest; validator confirmed current pool	✓ Accept	Continue same best pool
5	git-push-with-auth-fallback	Added "push hang treated as auth failure"; no improvement	✗ Reject	Keep current best pool
6	git-push-with-auth-fallback	Added identity config & filename consistency; no improvement	✗ Reject	Not admitted to next cycle

Accepted: 3 / 6 | Rejected: 2 / 6 | Retest: 1

Three Properties That Distinguish SkillClaw

01 Collective Evolution

Knowledge from individual interactions contributes to a shared, continuously improving skill ecosystem. No user's discovery is wasted.

02 Fully Automatic

Skill evolution is driven by runtime interaction without manual curation or explicit user intervention. The system improves itself.

03 Agentic Evolution Paradigm

Skill updates are produced through open-ended reasoning over evidence — not predefined update rules or fixed heuristics.

Evaluated on WildClawBench with Qwen3-Max • Compatible with OpenClaw, CoPaw, IronClaw, PicoClaw, ZeroClaw, NanoClaw, NemoClaw